

MACHINE LEARNING IN PRODUCTION

LECTURE 2 NOTES

YOUR DATA SCIENCE COULD USE SOME ENGINEERING



In this lecture we will cover

- 1) Difference between data scientist and machine learning engineer
- 2) Focusing on simplicity in all project work to reduce risk
- 3) applying agile fundamentals to ml project work
- 4) differences and similarities between DevOps and MLOps

In the last chapter, we looked at ML engineering through the lens of project work. But that's just one piece of the puzzle.

LET'S RECAP





- 1) Technology (tools, frameworks, algorithms)
- 2) People (collaborative work, communication)
- 3) Process (software development standards, experimentation rigor, Agile methodology)

This Lecture covers, at a high level, this trio of components for successful projects. Employing the appropriate balance of each, focused on creating maintainable solutions that are co-developed with internal customers in a collaborative and inclusive fashion, will greatly increase the chances of building ML solutions that endure.

primary focus of all DS work is to solve problems.

SOURCE:- MACHINE LEARNING IN ACTION, AUTHOR:- BEN WILSON

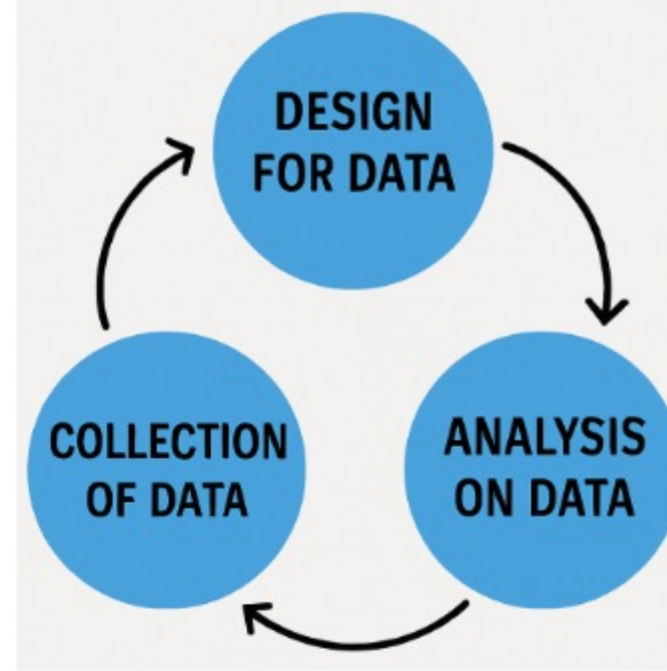
2.1) Augmenting a complex profession with processes to increase project success

In one of the earliest definitions of the term data science, as covered in Data Science, Classification, and Related Methods (Springer, 1996), compiled by C. Hayashi et al., the three main focuses are as follows:

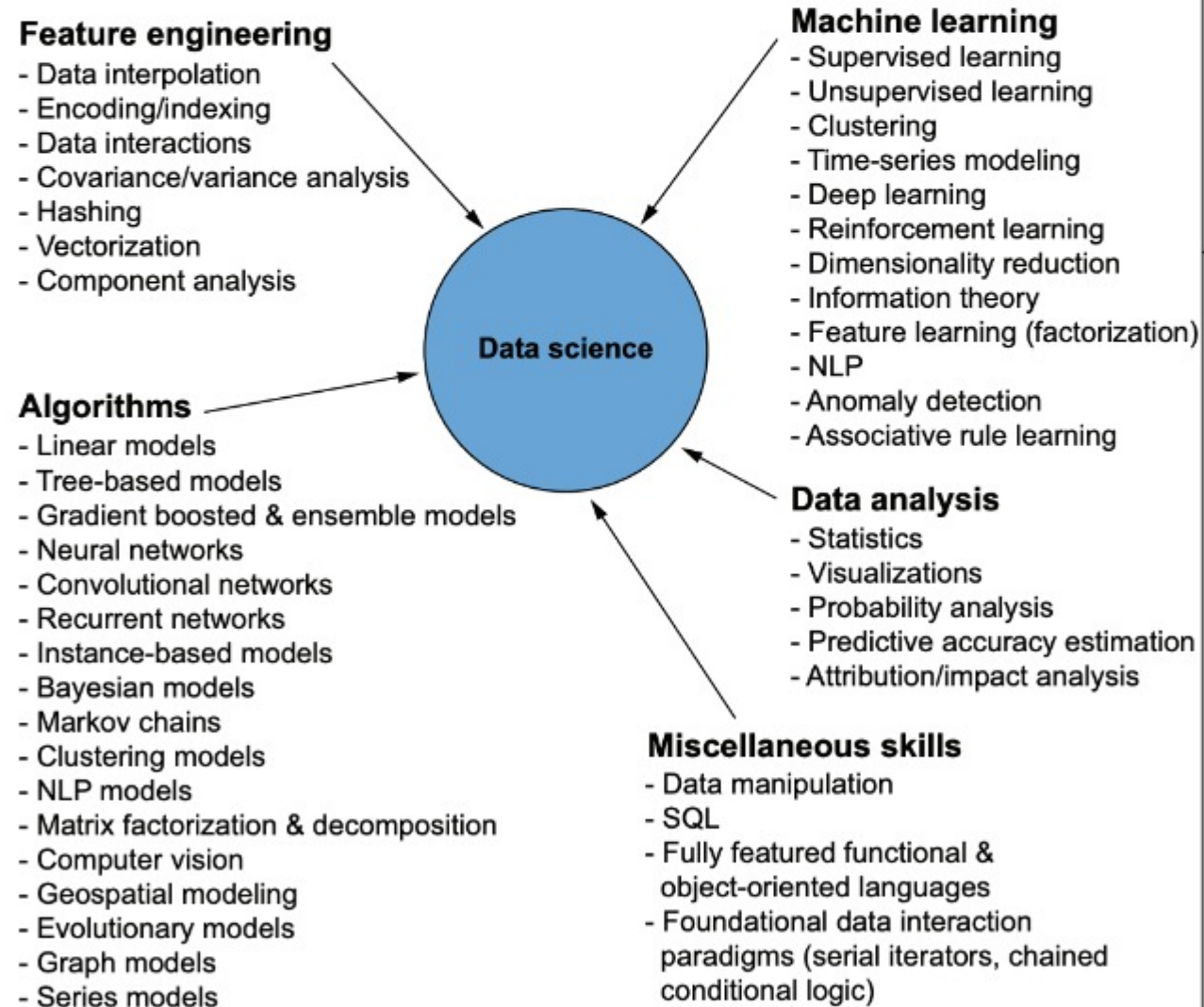
- 1) Design of Data
- 2) Collection of Data
- 3) Analysis of Data

Modern DS Roles focuses on last point and first two are handle by Data Engineers in some of the companies

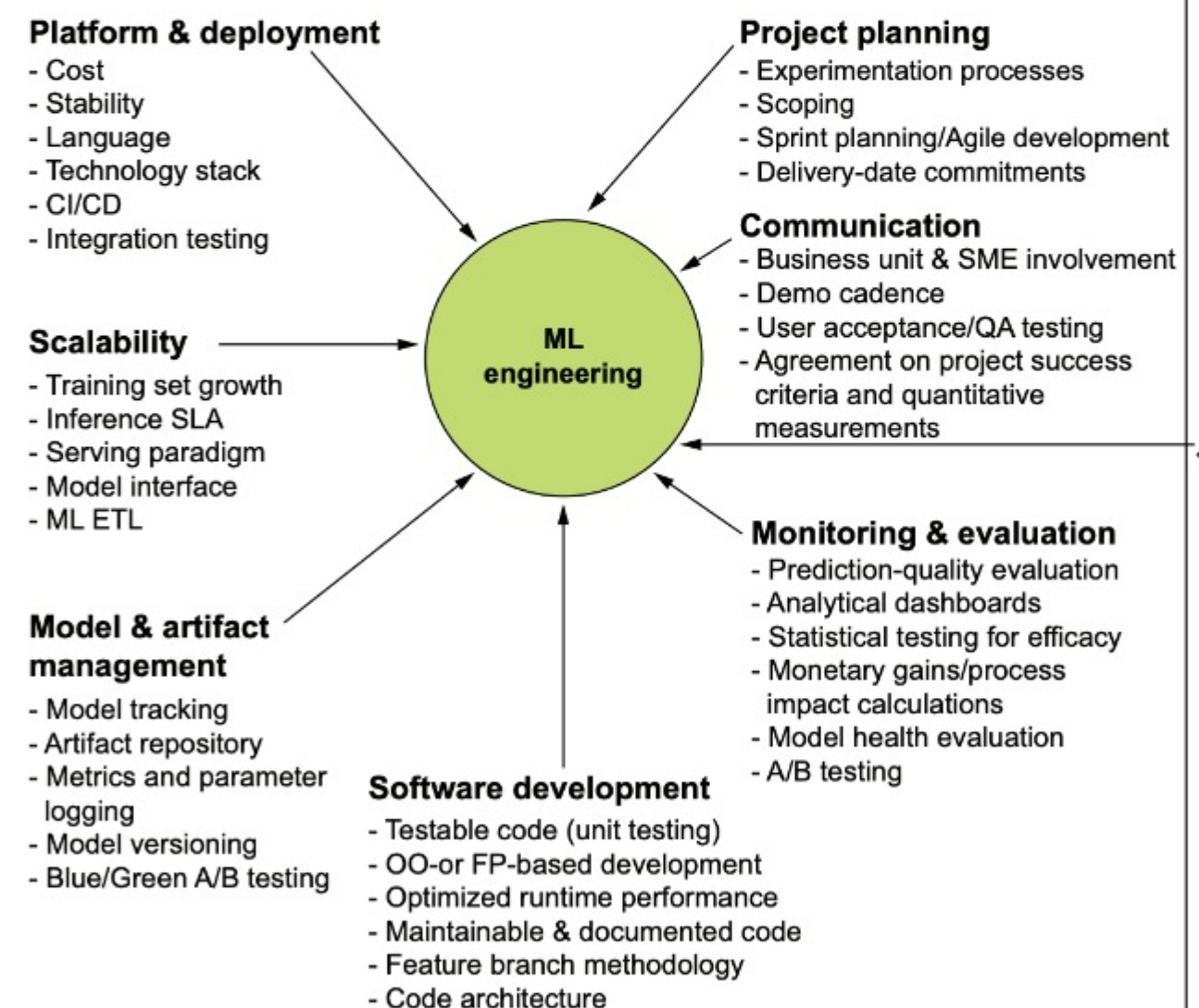




The (non-exhaustive) realm of data science



The (non-exhaustive) realm of ML engineering



SOURCE:- MACHINE LEARNING IN ACTION, AUTHOR:- BEN WILSON

ML engineering, as a concept, is a paradigm that helps practitioners focus on the only aspect of project work that truly matters: providing solutions to problems that actually work.



2.2) A Foundation of simplicity



Data scientists solve problems by applying mathematics to data creatively, without being tied to specific tools or algorithms.



Value comes from solving real problems, not from using trendy technology or the latest methods.



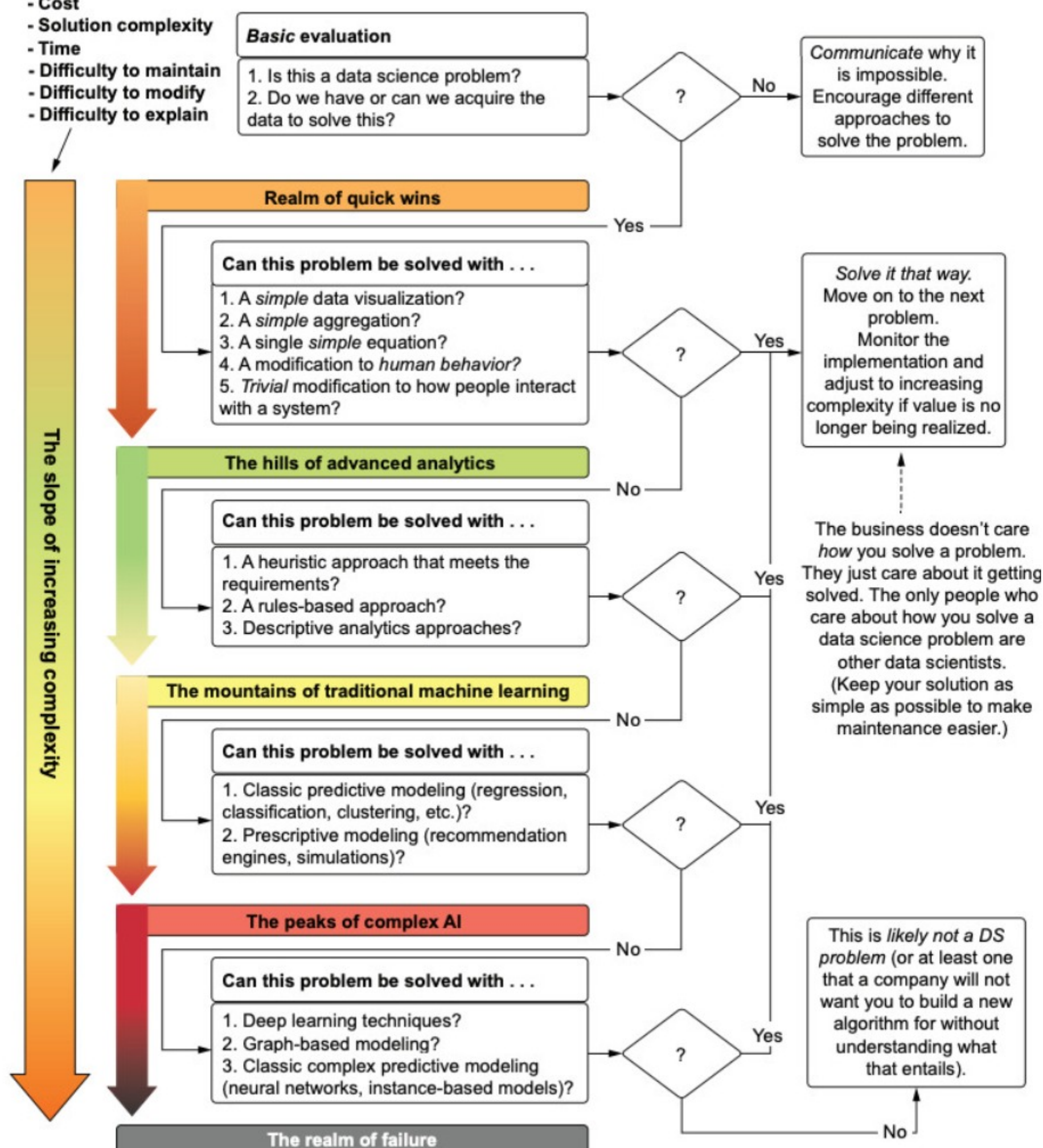
The best solutions are often the simplest ones that meet business needs effectively and efficiently.



This approach forms the core of ML engineering, guiding everything from project planning to execution and success.

Moving down this slope will cause increases in

- Cost
- Solution complexity
- Time
- Difficulty to maintain
- Difficulty to modify
- Difficulty to explain



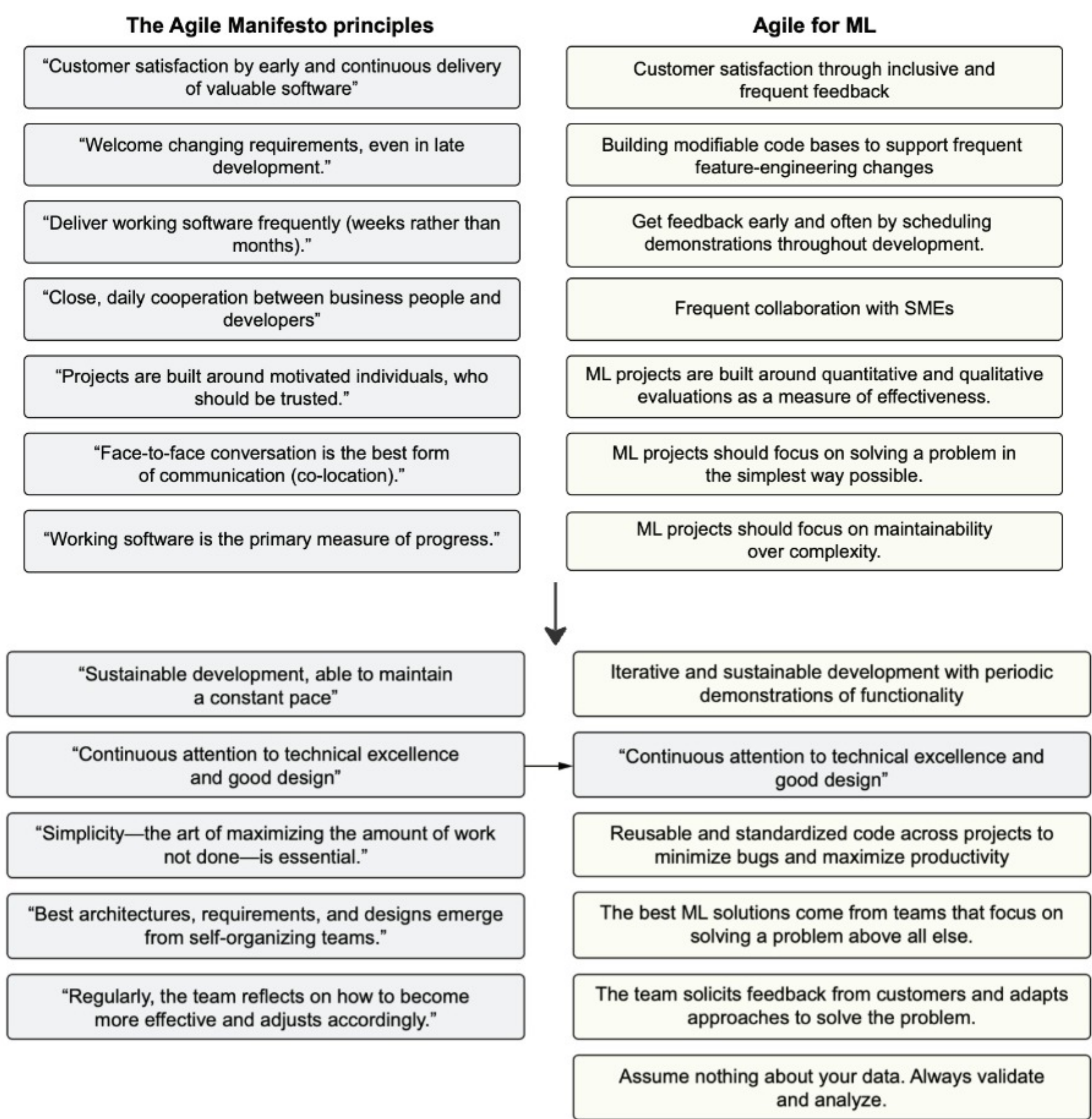
2.3) Co-opting principles of Agile software

What is meant by Agile Manifesto?



Author ben wilson and his colleagues took a stab at adapting these guiding principles to the field of data science

Let's relate the agile principles in software development with machine learning engineering



The Agile Manifesto is credited to the original team of 17 developers who met in Snowbird, Utah in 2001 to draft these principles, recorded in "Manifesto for Agile Software Development."

SOURCE:- MACHINE LEARNING IN ACTION, AUTHOR:- BEN WILSON

With this slight modification to the principles of Agile development, we have a base of rules for applying DS to business problems.

However, two critical points of Agile development can, when applied to ML project work, dramatically improve the way that a DS team approaches its work:

- 1)communication and cooperation
- 2)embracing and expecting change.

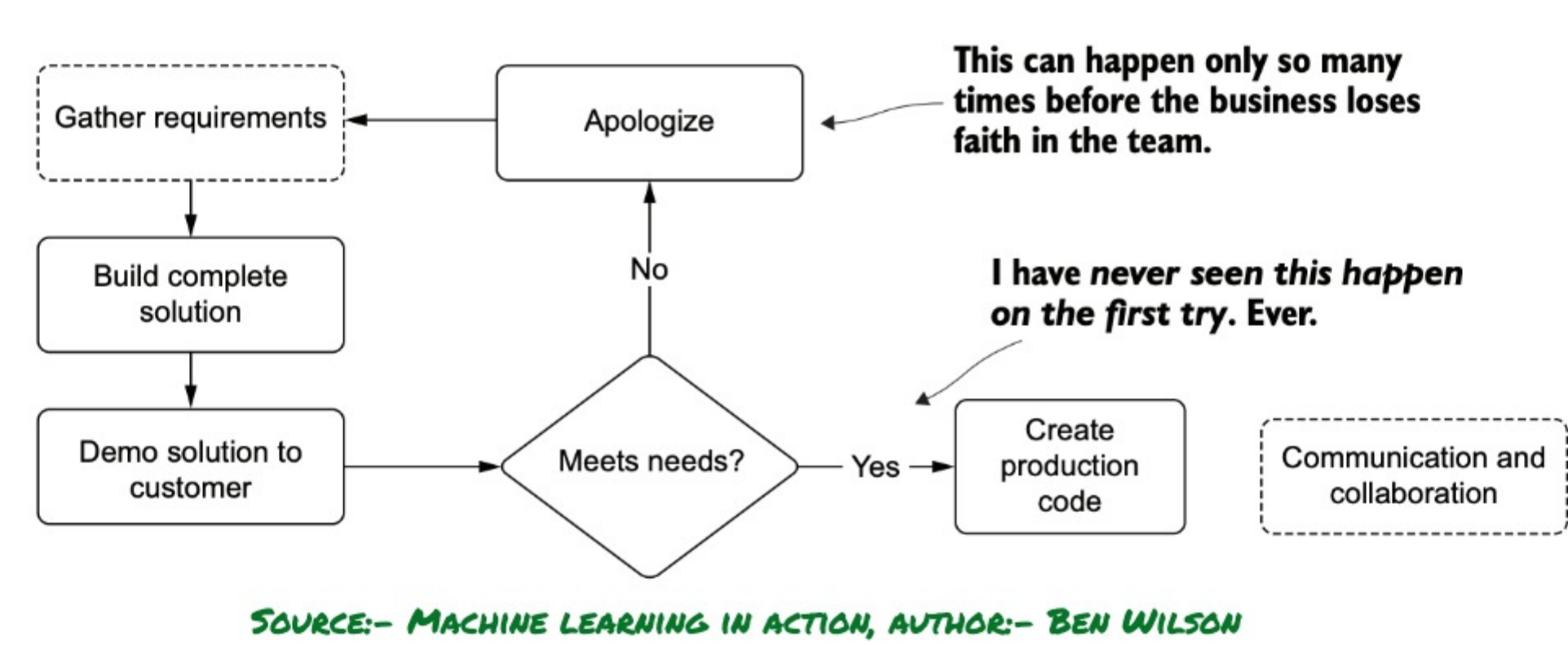
We'll take a look at these next

2.3.1) Communication and cooperation

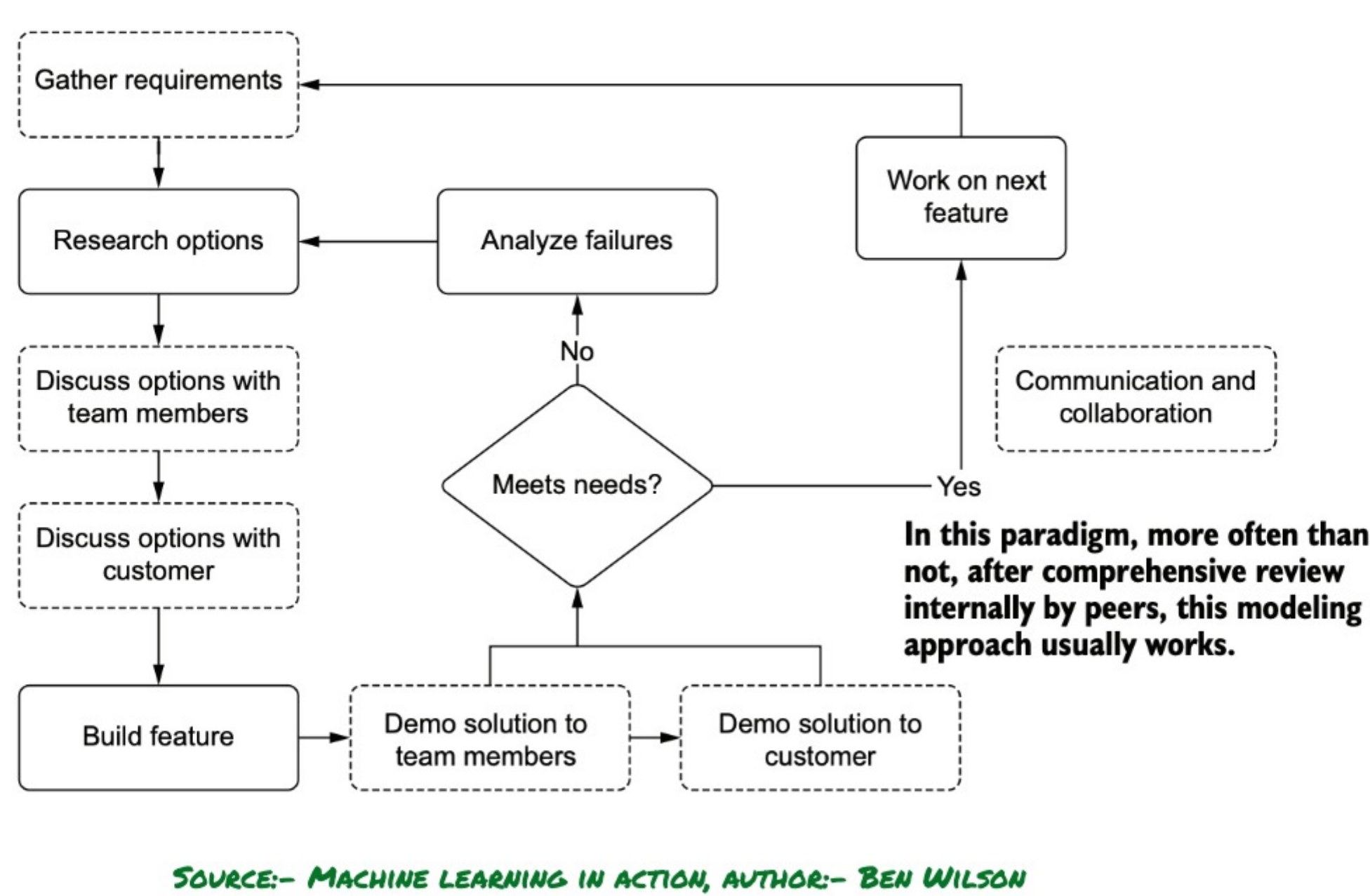
- 1) **Collaboration is even more vital in ML than in traditional Agile** because the work is complex and often unfamiliar to non-technical teams.
- 2) **Regular, clear discussions between ML teams and business units** help avoid misunderstandings and reduce costly rework.
- 3) **Strong communication within the ML team itself** is the second key to project success.
- 4) **Working solo may work in school, but not in real ML project** teamwork is essential for solving complex problems effectively.

Let us consider two scenarios:

1) Risky behaviour



2) AGILE-ALIGNED METHODOLOGY



Engaging more with your team leads to better ideas and stronger solutions. If you leave out your peers or business stakeholders, even in small decisions, there's a high chance the final outcome won't match their needs or expectations.

2.3.2) Embracing and expecting change

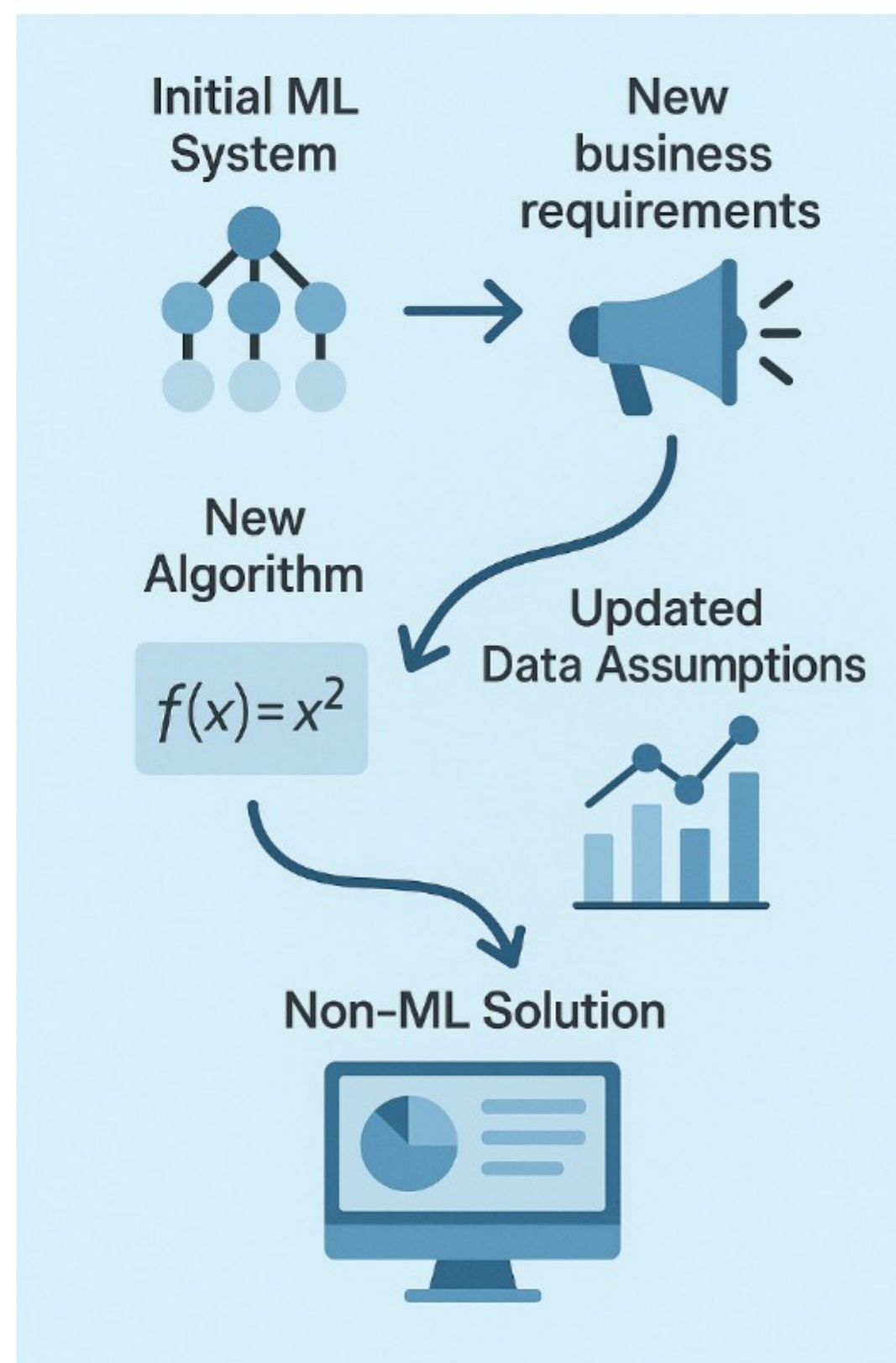
1)Change is inevitable in ML projects – from goals and algorithms to data assumptions and even the solution type (e.g., replacing ML with a simple dashboard).

2)Expecting change helps teams focus on the real goal – solving business problems, not chasing tools, trends, or fancy models.

3)Planning for change means designing modular, flexible systems – loosely coupled components make future updates much easier.

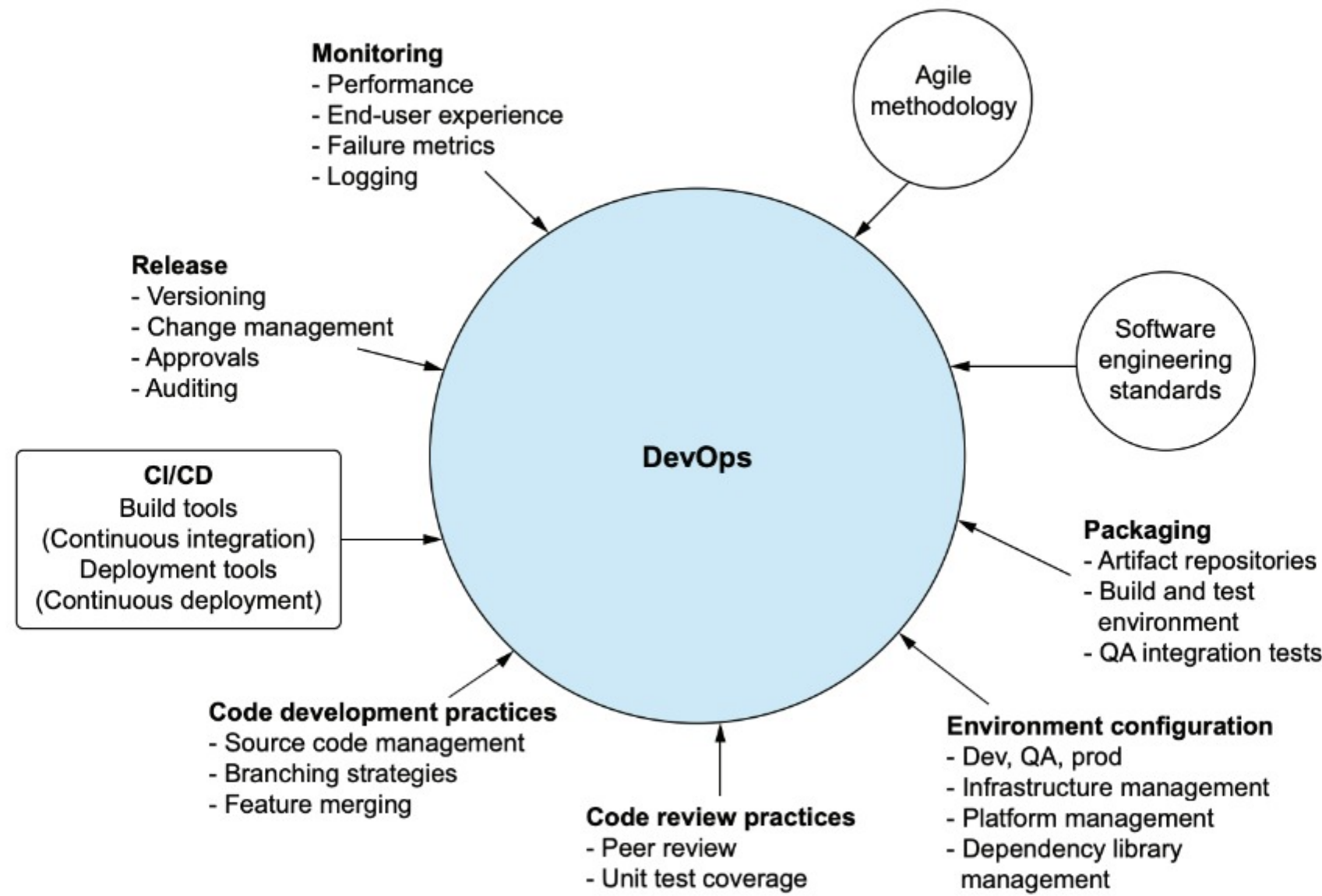
4)Agile methodology fits well with ML work – its iterative approach supports building adaptable, evolving solutions.

5)Thinking ahead saves effort – without this mindset, even small changes may require complete rewrites.



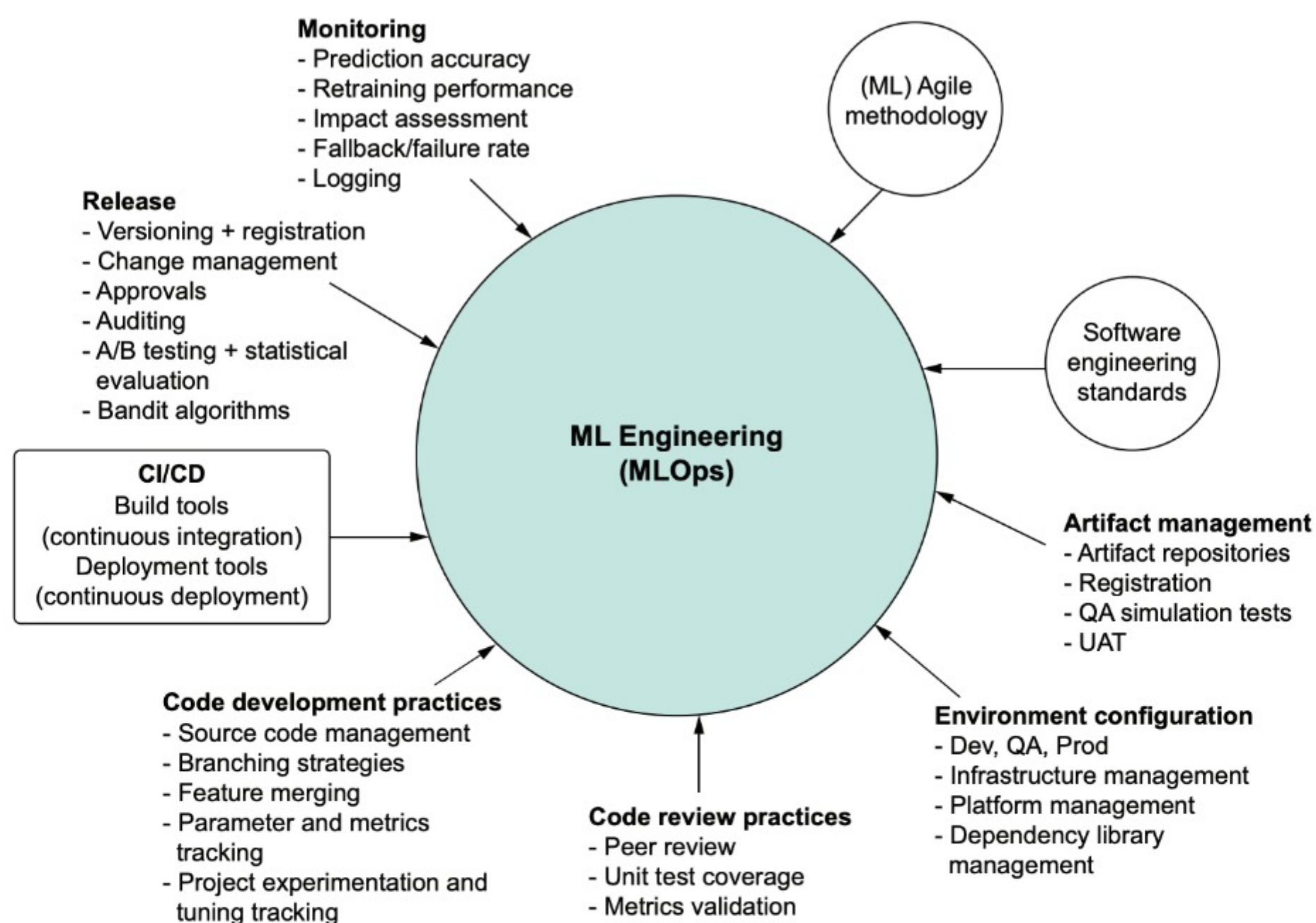
2.4) The foundation of ML engineering

Components of DevOps



SOURCE:- MACHINE LEARNING IN ACTION, AUTHOR:- BEN WILSON

Components of MLOps



SOURCE:- MACHINE LEARNING IN ACTION, AUTHOR:- BEN WILSON

In the upcoming lectures we will understand these components with real life example in detail

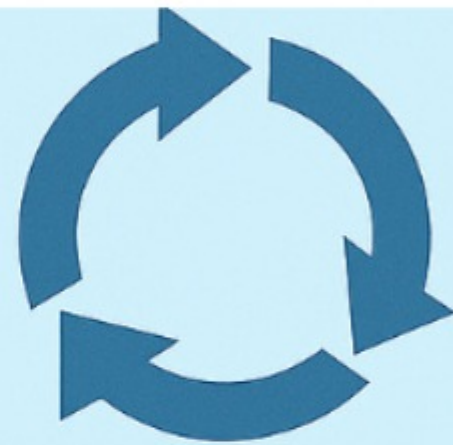


2..5) Summary



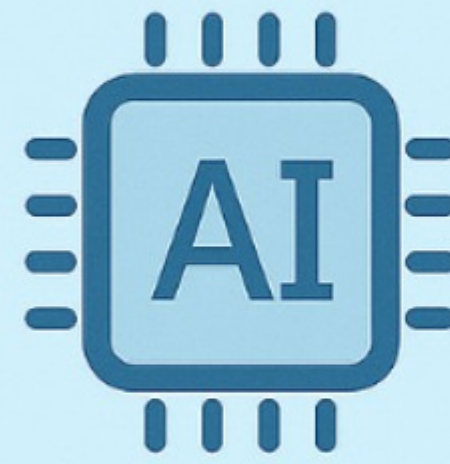
MLE engineers combine DS, DE, and SE disciplines to solve problems

Simpler solutions reduce development, computational, and operational costs



Agile scrum adapted to ML enables rapid, flexible, and testable development

MLOps extends DevOps with model management and continuous testing



In the next lecture we will discuss about planning and scoping of the project in detail.

THANK
YOU

